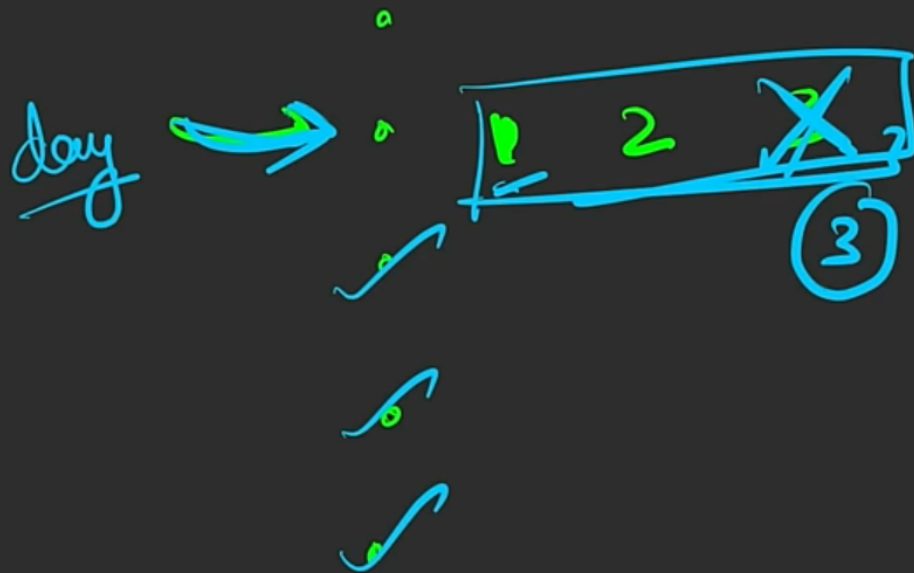


2.40



Top-down

$\sqrt{(n-1)} \rightarrow 0^{th}$

Recursion

1. $\sqrt{Incl}$, $last$
2. Do Steps
3. Main

$f(day, last)$



2.10

Quest

0 → task 0

1 → task 1

2 → task 2

3 → no task was done

(0-based
Indexing)

1.80

$f(n-1, 3)$



man merit points



$f(2, 1)$

man merit point



```

f(day, last)
{
    if (nd == 0)
    {
        mani = 0
        fun(i = 0 → 2)
        if (i != last)
            mani = mem(mani, desk[i])
        return mani;
    }
    mani = 0
    fun(i = 0 → 2)
    {
        if (i != last)
        {
            points = desk[day][i]
                    + f(day-1, i)
            mani = mem(mani, points);
        }
    }
    return mani;
}

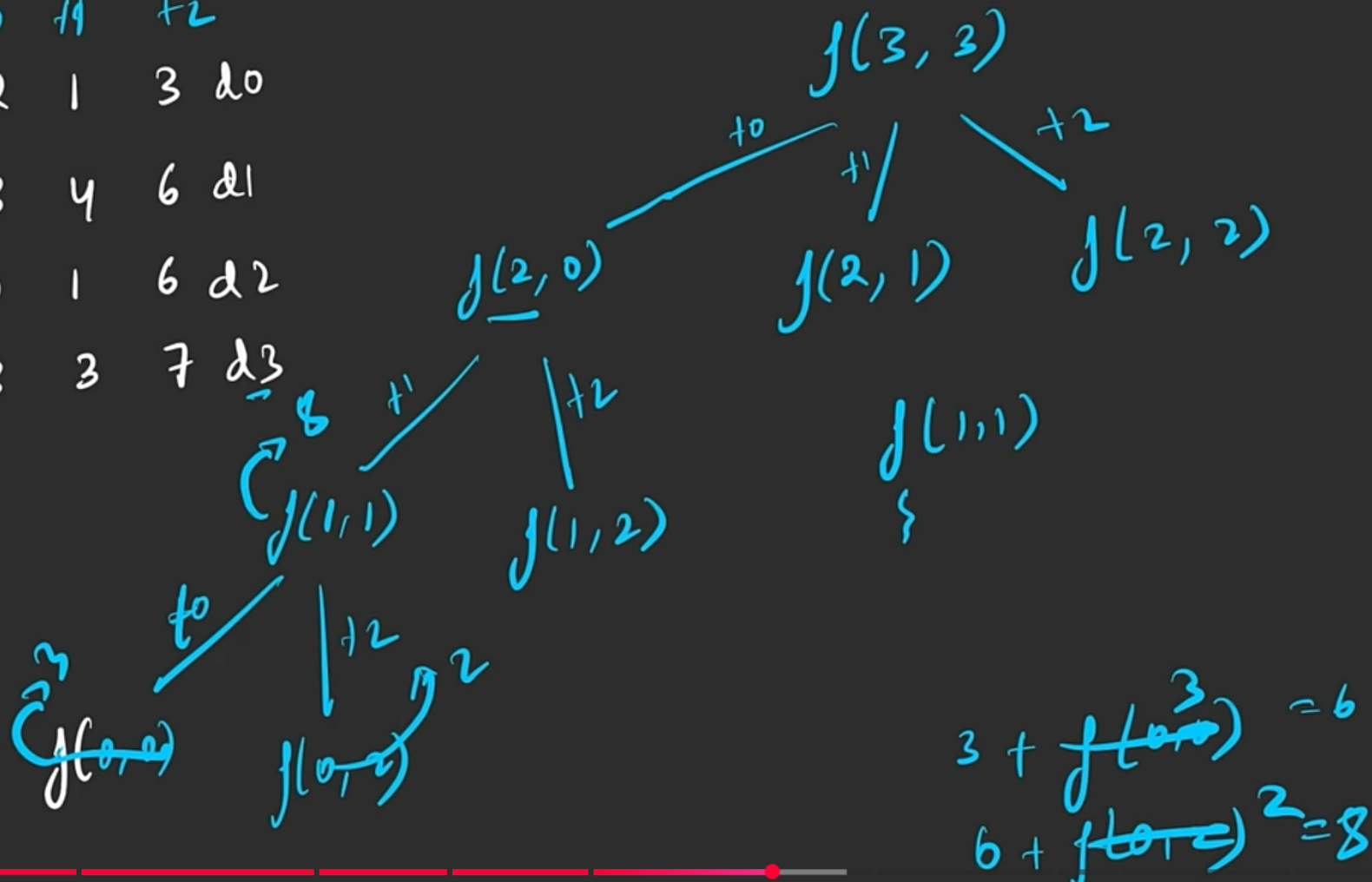
```

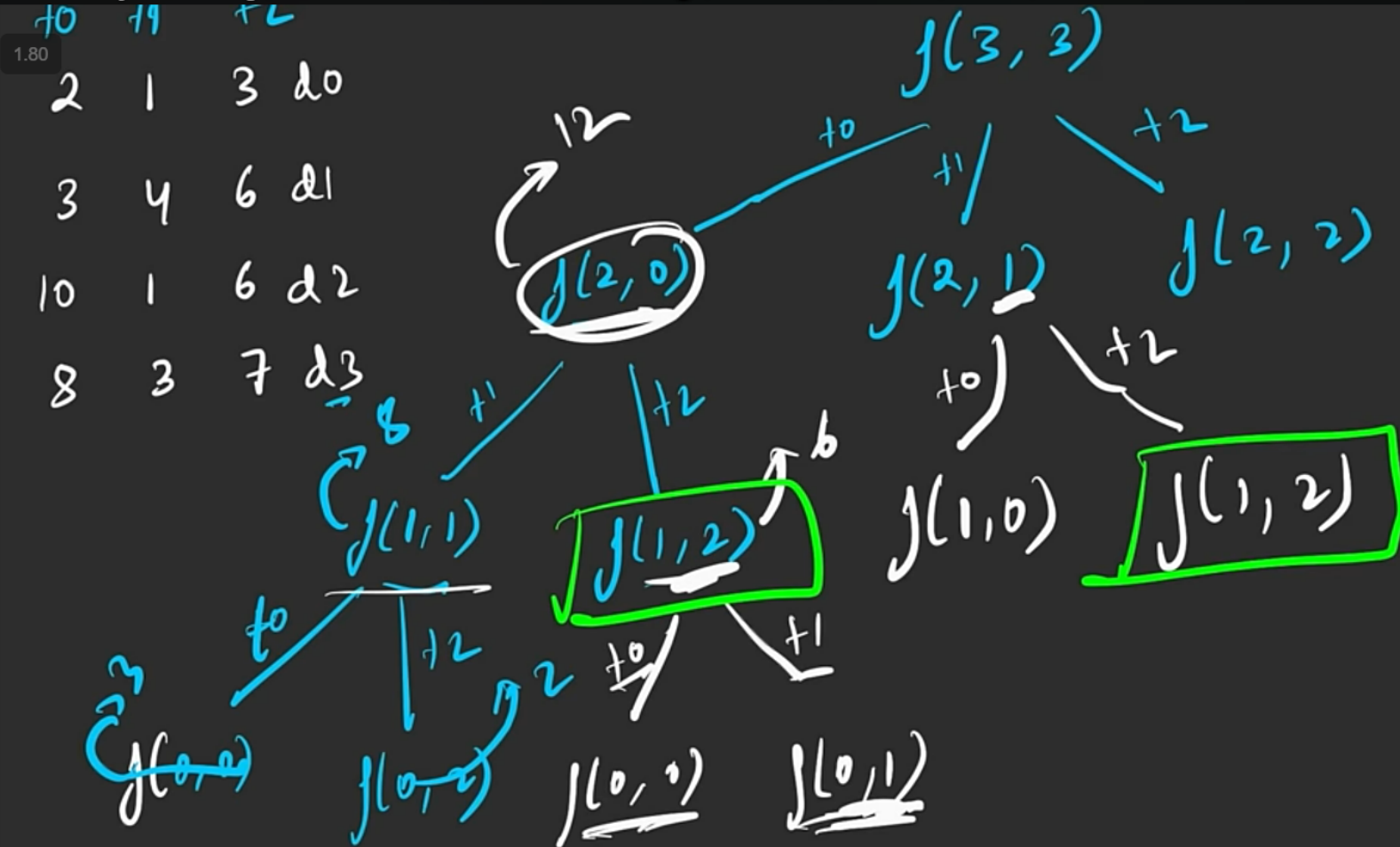


$n = 4$

1.80

t_0	t_1	t_2	
2	1	3	20
3	4	6	21
10	1	6	22
8	3	7	23





Problem

Submissions

For the first test case,
One of the answers can be:
On the first day, Ninja will learn new moves and earn 5 merit points.
On the second day, Ninja will do running and earn 3 merit points.
On the third day, Ninja will do fighting and earn 3 merit points.
The total merit point is 11 which is the maximum.
Hence, the answer is 11.

For the second test case:
One of the answers can be:
On the first day, Ninja will learn new moves and earn 70 merit points.
On the second day, Ninja will do fighting and earn 50 merit points.
On the third day, Ninja will learn new moves and earn 90 merit points.
The total merit point is 210 which is the maximum.
Hence, the answer is 210.

Sample Input 2:

```
2
3
18 11 19
4 13 7
1 8 13
2
10 50 1
```

```
1 int f(int day, int last, vector<vector<int>> &points) {
2     if(day == 0) {
3         int maxi = 0;
4         for(int task = 0; task < 3; task++) {
5             if(task != last) {
6                 maxi = max(maxi, points[0][task]);
7             }
8         }
9         return maxi;
10    }
11
12    int maxi = 0;
13
14    for(int task = 0; task < 3; task++) {
15        if(task != last) {
16            int point = points[day][task] + f(day-1, task, points);
17            maxi = max(maxi, point);
18        }
19    }
20    return maxi;
21 }
22 int ninjaTraining(int n, vector<vector<int>> &points)
23 {
24     return f(n-1, 3, points);
25     // Write your code here.
26 }
```

Previous

Next



Show Hint



Last saved on 3:51:00 AM



30:46 / 52:17

Memoization technique >



Correct Answer

! Report

Submitted on Jan 14, 2022, 3:52:33 AM

Penalty

Score ?

0%

100

Runtime

Language

319ms

C++ (g++ 5.4)

Previous Submissions

Submitted On	Status	Score	Penalty
few secs ago	Correct Answer	100	0%
2 hours ago	Correct Answer	100	0%
2 hours ago	Correct Answer	100	0%
2 hours ago	Correct Answer	100	0%
2 hours ago	Time Limit Exceeded	18	0%

```
20     }
21     }
22     return dp[day][last] = maxi;
23 }
24 int ninjaTraining(int n, vector<vector<int>> &points)
25 {
26     vector<vector<int>> dp(n, vector<int>(4, 0));
27
28     dp[0][0] = max(points[0][1], points[0][2]);
29     dp[0][1] = max(points[0][0], points[0][2]);
30     dp[0][2] = max(points[0][0], points[0][1]);
31     dp[0][3] = max(points[0][1], max(points[0][1], points[0][2]));
32
33     for(int day = 1; day < n; day++) {
34         for(int last = 0; last < 4; last++) {
35             dp[day][last] = 0;
36
37             for(int task = 0; task < 3; task++) {
38                 if(task != last) {
39                     int point = points[day][task] + dp[day-1][task];
40                     dp[day][last] = max(dp[day][last], point);
41                 }
42             }
43         }
44     }
45
46     return dp[n-1][3];
```

1.80

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Problem Submissions

Correct Answer

Submitted on Jan 14, 2022, 4:09:31 AM

Penalty 0% Score 100

Runtime 329ms Language C++ (g++ 5.4)

Previous Submissions

Submitted On	Status	Score	Penalty
few secs ago	Correct Answer	100	0%
16 mins ago	Correct Answer	100	0%
2 hours ago	Correct Answer	100	0%
2 hours ago	Correct Answer	100	0%
2 hours ago	Correct Answer	100	0%
2 hours ago	Time Limit Exceeded	18	0%

```
24 int ninjaTraining(int n, vector<vector<int>> &points)
25 {
26     vector<int> prev(4, 0);
27
28     prev[0] = max(points[0][1], points[0][2]);
29     prev[1] = max(points[0][0], points[0][2]);
30     prev[2] = max(points[0][0], points[0][1]);
31     prev[3] = max(points[0][1], max(points[0][1], points[0][2]));
32
33     for(int day = 1; day < n; day++) {
34         vector<int> temp(4, 0);
35         for(int last = 0; last < 4; last++) {
36             temp[last] = 0;
37
38             for(int task = 0; task < 3; task++) {
39                 if(task != last) {
40                     temp[last] = max(temp[last], points[day][task] + prev[task]);
41                 }
42             }
43         }
44         prev = temp;
45     }
46
47     return prev[3];
48
49     // Write your code here.
50 }
```

Run Code



1.80

